6 October 1981

MILITARY SPECIFICATION

PROPELLANT, IGNITER, X-62

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for one type of igniter reduced smoke propellant. The propellant is composed essentially of a dispersion of ammonium perchlorate oxidizer in a matrix of synthetic elastomer which serves the dual function of fuel and binder.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-T-100	Tape, Pressure-Sensitive Adhesive, Polyester Film.

MILITARY

MIL-D-3464	Desiccant, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-A-82667	Ammonium Perchlorate, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1420

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MIL-I-85492	Imine Curing/Bonding Agents.
MIL-C-85493	Catalyst, Burn-Rate 2, 2-Bis (Ethyl-ferrocenyl Propane).
MIL-D-85494	Dioctyl Adipate.
MIL-B-85497	Hydroxyl-Terminated Polybutadiene.
MIL-C-85498	Curing Agent, Dimeryl-diisocyanate and Isophorone diisocyanate.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-2126	Reduced Smoke Rocket Motor Processing and Test Procedures.
JANNAF CPIA 21	Solid Propellant Mechanical Behavior Manual.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

STANDARDS

American Society for Testing and Materials (ASTM)

ASTM-E-203	Standard Test Method for Water Using Kark Fischer Reagent.
ASTM-D-1348	Standard Test Method for Moisture in Cellulose.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

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3. REQUIREMENTS.

3.1 Composition. The nominal composition of the propellant shall be in accordance with Table I.

3.1.1 Physical property control.

3.1.1.1 Polymer-curing agent ratio. The ratios of hydroxyl-terminated polybutadiene (HTPB) polymer and isocyanate curing agent may be varied as a means of optimizing and controlling cured propellant tensile properties. Acceptable ratios shall be between 0.80 and 1.00 equivalents isocyanate per equivalent hydroxyl in the HTPB polymer.

3.1.2 Ballistic property control.

3.1.2.1 Ammonium perchlorate. The ratio of ground to unground ammonium perchlorate may be varied from a nominal ratio of 50:50 to control ballistic properties.

3.1.2.2 Burn-rate catalyst. The concentration of catalyst may be varied as a means of adjusting and controlling propellant ballistic properties. Catalyst content may vary from 1.0 to 2.0 percent, with compensating adjustment in the total HTPB polymer and curing-agent content.

3.2 Weighing accuracy. Each propellant ingredient shall be weighed to within 0.3 of one percent of the weight requirements calculated from the weight percent specified.

3.3 Manufacturing process. The propellant shall be mixed, cast, and prepared as specified in MIL-STD-2126, Method II.

3.4 Physical properties. The physical properties of the cured propellant, when tested in accordance with 4.3 at 25 ± 3 degrees Celsius (^{0}C) and at a crosshead speed of 2 inches per minute shall conform to Table II.

3.5 Ballistic properties. The propellant strands shall demonstrate an average burn rate of 1.80 ± 0.15 inches per second when tested at 38 + 2^{0}C and at 1000 pound-force per square-inch gage (psig). Testing shall be in accordance with 4.4.

3.6 Toxic products and safety. Safety regulations and guidelines applicable to the use of propellant should be complied with to preclude personal injury and damage to equipment and facilities.

3.7 Workmanship. Workmanship shall be such that the propellant is uniform in appearance, of consistent high quality and free from visible contamination.

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TABLE I. Nominal propellant composition.

Ingredient	Document	Function	Nominal percent by weight 4/
Hydroxyl-terminated polybutadiene	MIL-H-85497, Type II, Class 2	Binder and fuel	13.263
3-isocyanatomethyl-3, 5, 5-trimethylcyclohexyl-isocyanate	MIL-C-85498, Type II	Curing agent	0.937
Dioctyl adipate	MIL-D-85494	Plasticizer	1.00
1, 1'-(1,3-phenylene dicarbonyl) bis [2-methyl-aziridine]	MIL-I-85492, Type I	Bonding agent	0.30
Catalyst, burn-rate, 2, 2-bis (ethylferrocenyl) propane	MIL-C-85493	Burn rate catalyst	1.50 1/
Ammonium perchlorate	MIL-A-82667/2 Type I, Class 1 and Ground 2/	Oxidizer Oxidizer	83.0 3/

1/ See 3.1.2.2.

2/ Type I, Class 1 ammonium perchlorate shall be ground to a particle size (weight mean diameter, microns) of 2.0 ±0.4 microns (see 4.5). After grinding, the external moisture shall be no greater than 0.03 percent (see 4.6).

3/ See 3.1.2.1.

4/ See 3.2.

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TABLE II. Physical properties. 1/

Parameter	Minimum
Strain at maximum stress, in/in	0.10
Maximum stress, pound-force per square inch (psi)	100

1/ Physical properties shown are corrected values (see 6.4).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Samples shall be taken from each propellant mix as required for physical and ballistic-property testing. A minimum of three specimens shall be tested for physical properties, and a minimum of six strands shall be tested for ballistic properties from each mix of propellant. Specimens shall be prepared in accordance with MIL-STD-2126, Method II.

4.3 Physical properties. The cured propellant tensile properties shall be determined in accordance with JANNAF Solid Propellant Mechanical Behavior Manual (CPIA Publication 21) using class B or C specimens. Sample preparation and data evaluation shall be in accordance with CPIA Publication 21.

4.3.1 Specimen conditioning. Tensile specimens shall be conditioned for a minimum of 1 hour at 25 ± 3°C. The relative humidity of the testing environment shall not exceed 50 percent. When not in specified conditioning, the specimens shall be stored at 25 ± 3°C and a relative humidity not exceeding 50 percent.

4.4 Ballistic properties. Burn-rate specimens shall be prepared and inhibited in accordance with MIL-STD-2126, Method II.

4.4.1 Time recorder. The time recording device shall be capable of measuring burning time intervals within one percent for each 3-inch increment between fuze wires.

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4.4.2 Conditioning. Strands shall be stored at 38 ± 2°C for a minimum of 60 minutes prior to testing.

4,4.3 Testing. The strands shall be tested in a clean strand burner conditioned to 38 + 2°C and regulated to an initial operating pressure of 1,000 + 10 psig.

4.4.4 Data. Strands shall be burned recording the pressure and the burning time between fuze wires.

4.4.5 Calculations. The following calculations shall be performed to determine the average burning rate at 1,000 psig.

Calculations:

$$r_i = \frac{L}{t_i}$$

$$\overline{r} = \frac{r_1 + r_2 + r_3 + r_4 + r_5 + r_6}{6}$$

Where:

r_1 = burning rate for strand number i

L = distance between fuze wires (3.000 inches)

t_i = burning time for strand number i

$\overline{r}$ = average burning rate for six acceptable strands

4.5 Particle size (weight mean diameter).

4.5.1 Apparatus. Particle size analysis equipment (Mine Safety Appliances (MSA) Company, Pittsburgh, Pennsylvania, or equal).

4.5.2 Reagents.

a. Settling liquid - Reagent grade chlorobenzene.

b. Feed liquid - volumetric mixture of 60 percent chlorobenzene and 40 percent benzene (reagent grades).

c. Surfactant - Twitchell Base 8240 (Emery Industries, Inc., Cincinnati, Ohio), or equal.

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4.5.3 Procedure. Weigh a 0.012 gram sample onto a 4- by 4-inch glass plate. Add one drop of Twitchell Base 8240 surfactant, or equal. Paste the sample for 2 minutes with a steel spatula. The resultant paste should be a uniform dispersion. Transfer the paste to the MSA feed chamber by means of a sharpened steel spatula. Add approximately 0.8 to 1.0 milliliter (ml) of feed liquid. Hold the chamber between the thumb and index finger and shake to bring the sample into suspension. Introduce the sample into the 1.0-millimeter (mm) MSA tube (prefilled to the mark with chlorobenzene) and start the analysis schedule. Determine the weight mean diameter, and report value to the nearest 0.1 micron.

4.6 External moisture. The Karl Fischer (KF) method shall be used with any suitable apparatus. The general procedure for total moisture content shall be by titration of a solution of the sample in a 3:1 mixture of pyridine and methanol. The surface moisture is determined by titration of a slurry of the solid sample in a solution or liquid medium saturated with ammonium perchlorate. The end point may be determined either visually or electrometrically by means of either a direct or back titration of a measured excess of KF reagent. The direct titration shall be in accordance with ASTM-E-203 and the back titration shall be in accordance with ASTM-D-1348. The standardization of the KF reagent shall be performed daily in accordance with ASTM-E-203. All the standard precautions shall be taken to protect the specimens against change in moisture content and to protect all reagents, solvents and the titration mixture against the entry of water. During all the titrations, the stirring action shall be at the highest speed which will not cause splashing or bubble formation.

5. PACKAGING.

5.1 Packing and storage. Strands shall be packed in conductive plastic with no more than one pound of strands to a container. Strands shall be packed parallel and straight. Precaution shall be taken to prevent bending or deforming of strands. The plastic shall be sealed with moisture-proof tape conforming to L-T-100 and placed in moisture-proof containers with a one-third unit (or larger) of desiccant conforming to MIL-D-3464, Type I. The lid of the container shall either be moisture proof or sealed with moisture-proof tape. The sealed containers shall be stored in the temperature range of 5 to 32°C.

5.2 Marking. Marking shall be in accordance with MIL-STD-129 and shall include the following information:

a. Propellant designation

b. Mix number

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c. Specification number

d. Part number

e. Number of parts

f. Date packed

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The propellant is intended for use in the MARK 297 Igniter.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Responsibility for inspection and inspection facilities, if different than 4.1.

6.3. Definitions.

6.3.1 Lot. A lot consists of one batch of propellant (see 6.3.2). At place of delivery, a lot consists of propellant from one supplier's lot received in a single shipment. Partial shipments may be considered as a single shipment by the procuring activity.

6.3.2 Batch. A batch consists of propellant made as one unit in a single mixer using an unchanged manufacturing process.

6.4 Physical properties correction method. The physical properties correction method used by the contractor should be the procuring activity for concurrence.

Preparing activity:
Navy - AS

(Project 1420-N244)

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